

Genetic food intolerances (626 foods)

Genetic predisposition predicted for 626 frequent consumption foods worldwide, by category
Report generated at 2 of October 2026 (local time)

Genetic basis: Raw DNA of a genotyping chip (GRCh38, 684 967 SNPs) + local imputation (Beagle 5.5, full genome reference panel 30x; only genotypes with DR2 ≥ 0.8). Genetic data did not leave the local computer. Only genetics: no analyses or exams were used.

Mechanisms evaluated: 28 mechanisms (lactase, celiac disease, hereditary intolerance to fructose, sucrase-isomaltase, histamine, caffeine, salt, alcohol, liver x sugar, glycemic response, iron, smoking detoxification, favism, oxalate, purines, PKU, saturated fat, bitter taste and coriander, among others), with ~140 variants and ClinVar sweep of ~130 food metabolism genes.

List of foods: Foods most consumed worldwide (Food Sheets products from FAO unfolded in concrete foods, food composition tables and frequent processed products), without duplicates.

Important: is a genetic reading of predisposition, not a diagnosis. Genetics explains only one part of the intolerances; IgE allergies and FODMAP sensitivity are not predictable by genes. All indications are suggestion (not applied).

red avoid / high probability **orange** limit, dose-dependent effect **yellow** monitoring / possible sensitivity. On request, the report shows only predisposed foods; the rest are counted but not listed.

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1. Executive summary

Class	Food No.	%
Avoiding high probability of intolerance	0	0.0%
Limit (dose dependent)	140	22.4%
Monitoring / possible sensitivity	162	25.9%
No genetic predisposition (gene evaluated without alteration, non-determinable or no known genetic binding)	324	51.8%
Total analyzed	626	100%

324 of the 626 verified foods do not show genetic predisposition (are not listed in this report).

Fast reading. No food is red: there is no genetic intolerance "classical". Orange and yellow colors come from dose-dependent effects, not true intolerance:

- **Free sugar and excess fructose** (Orange; sugar, honey, syrups, soft drinks, juices, sweets and pastries): PNPLA3 I148M + MBOAT7 + Pi*MZ probably make the liver more sensitive to high doses of sugar; the protective HSD17B13 attenuates. The whole fruit does not enter here.
- **Sal** (Orange in very salty: sausages, ham, very salty cheeses, salty candies, soy sauce, cube juices, snacks): AGT, ADD1, CYP11B2, GNB3 and ACE in salt sensitivity profile.
- **Coffee** (Orange in coffee and energy drinks; yellow in tea, glue and black chocolate): CYP1A2 *1F A/C (normal to intermediate elimination) + ADORA2A C/C (less sensitive); the limit comes from sleep (insomnia p80).
- **Alcohol** (Orange in all alcoholic beverages): no flushing (ALDH2 and ADH1B normal), but the genetic liver (PNPLA3 + Pi*MZ) increases susceptibility.
- **Histamine** (yellow: cured cheeses, sausages, fish in canned or smoked, fermented, wine, vinegar, spinach, eggplant, avocado): DAO with 2 heterozygous variants + HNMT Thr105Ile heterozygous. Weak evidence.
- **Sucrose** (yellow; reinforces sweet orange): SI Val15Phe heterozygous, with slightly lower sucrase activity.
- **Refined hydrates** (yellow: white bread, white rice, potato, refined flour): TCF7L2, FTO, MTNR1B and FABP2 heterozygotes — intermediate glycemic response, worst at night.
- **Other yellows:** viscera and morcella (ferro heme; HFE H63D heterozygous), carbonized and smoked meat (GSTM1 probable null + GSTA1*B/*B + NAT2 slow), fortified foods (iron and folic acid; HFE + compound MTHFR), liquor (salt sensitivity) and coriander (OR6A2: only possible soap flavor).

Good genetic news: Lactose tolerant (LCT, 2 SNPs), without genetic risk of celiac disease (without DQ2/DQ8), without hereditary intolerance to fructose (ALDOB), without favism (G6PD), without PKU (PAH), typical response to saturated fat (APOA2, APOE ε3/ε3, LIPC), low predisposition to high uric acid (normal ABCG2), without genetic hyperoxaluria (AGXT), intermediate taster of bitter in the crucifers (which with null GSTM1 give even more benefit) and secretor FUT2.

What this work adds to what was already known

- **Sucrase-isomaltase (SI) Val15Phe heterozygous** (rs9290264 C/A, chip): a common variant of reduced function associated with irritable bowel syndrome and symptoms with sucrose. No G1073D. New.
- **ALDOB:** A149P, A174D and N334K absent → hereditary intolerance to fructose very unlikely. New.
- **PAH:** 9 frequent variants of PKU absent (before only R408W). New.
- **Lactase confirmed by 2.0 SNP** (−22018, rs182549) and without other persistence alleles. DQ2.2 also absent. Reinforcement.
- **Sal:** The high sensitivity profile of the previous interpretation is explained by ADD1 Gly460Trp, CYP11B2 −344 T/T, GNB3 825T and AGT M235T. Fine new.

- **Caffeine:** rs2472297 and rs2470893 C/C (CYP1A1/CYP1A2, without the fast elimination allele) are added to CYP1A2 *1F A/C and ADORA2A C/C. Fine new.
- **Coriander:** OR6A2 rs72921001 C/A, 1 allele associated with soap flavor (small effect). New.
- **AGXT** Heterozygous haplotype (P11L + I340M) without G170R: no isolated clinical relevance. **APOA2** –265 not C/C and **LIPC** –514 C/C: typical response to saturated fat. New.
- **ClinVar in ~130 genes of food metabolism** (chip + imputation): no new pathogenic variant; only Pi*Z already known. New.
- Confirmed: DAO/HNMT slightly reduced, G6PD normal, ALDH2/ADH1B normal, FUT2 secretory, GALT Duarte heterozygote (without relevance), HFE H63D heterozygote, MTHFR compound, GSTM1 null probable.

Non-genetic notes (do not enter the colors): Grapefruit and pomelium interact with drugs metabolized by CYP3A4; unfiltered coffee (embolus, boiled) rises LDL by the coffee; vitamin K of green leaves only matters if one day there is anticoagulation with coumarins (VKORC1 T/T + CYP2C9 *1/*3).

2. Mechanisms and genes evaluated

Mechanism	Genes	Genotype (strand +, GRCh38)	Interpretation	Evidence	Trust
Free sugar/fructose x liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	PNPLA3 I148M C/G; MBOAT7 C/T; Probable Pi*MZ; HSD17B13 protector (attenuous)	High doses of free sugar/fructose increase liver fat in patients with 148M	Moderate	chip + imputation
Coffee	CYP1A2, ADORA2A	CYP1A2 *1F A/C (normal to intermediate elimination; rs2472297 and rs2470893 C/C, without the fast allele) + ADORA2A rs5751876 C/C	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p80): dose and time dependent effect	Moderate	chip
Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	AGT M235T A/G; ADD1 G460W G/T; CYP11B2 -344 T/T; GNB3 825 C/T; ACE (rs4343 G/A)	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Moderate-weak	chip
Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	ALDH2 G/G and ADH1B C/C (without flushing); PNPLA3 I148M; Probable Pi*MZ; MBOAT7 C/T	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Moderate-strong	chip + imputation
Sucrose (sucrose-isomaltase)	SI	Val15Phe (rs9290264 C/A) heterozygous; without G1073D	Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate-weak	chip
Food Histamine	AOC1 (DAO), HNMT	DAO Thr16Met C/T + His645Asp C/G; HNMT Thr105Ile C/T (all heterozygous); without -691T or Ser332Phe	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Weak	chip
Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	TCF7L2 rs7903146 C/T + rs12255372 G/T; FTO het; MTNR1B C/G; FABP2 Ala54Thr het	Intermediate glycemic response; worst at night (MTNR1B)	Moderate-weak	chip + imputation
Iron heme	HFE, TMPRSS6	HFE H63D C/G (without C282Y or S65C); TMPRSS6 A/G	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Weak	chip
Fortified foods (iron, folic acid)	HFE, MTHFR	HFE H63D C/G; MTHFR C677T G/A + A1298C T/G	Iron added and synthetic folic acid: prefer natural sources	Weak	chip
Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Probable null GSTM1; GSTA1 *B/*B; NAT2 *5/*5 (slow)	Weakest II phase detoxification for HAP and heterocyclic amines	Weakness	chip + inference (GSTM1 is CNV)
Coriander (soap flavor)	OR6A2	rs72921001 C/A (1 allele "soap")	A slightly higher probability of finding the coriander with flavor (only flavor)	Weak	chip
Liquorice (glycyrrhizal)	CYP11B2, AGT, ADD1	CYP11B2 -344 T/T + salt sensitivity profile	Glycyrrhizin retains sodium; more relevant with salt sensitivity	Weak	chip
Lactose	LCT/MCM6	-13910 C/T (rs4988235 G/A) + -22018 G/A (rs182549 C/T): 1 persistence allele	Lactase persistence (dominant): lactose tolerant	Strong	chip (2 SNPs in agreement)
Gluten — Celiac Disease	HLA-DQ2.5/DQ2.2/DQ8	rs2187668 C/C; rs7454108 T/T; negative DQ2.2 tags; HLA imputed without DQ2/DQ8; PGS celiac p3.9	Without the haplotypes required for celiac disease (negative predictive value ≈ 99%)	Strong	chip + imputation
Hereditary intolerance to fructose	ALDOB	A149P, A174D and N334K absent	Without the variants that cause most cases	Strong	chip
Favism	G6PD	Without A-, without c.563C>T (normal hemizygous)	No genetic risk of hemolysis with fava beans	Strong	chip
Saturated fat	APOA2, APOE, LIPC	APOA2 -265 T/C (not C/C); APOE ε3/ε3; LIPC -514 C/C	Typical response to saturated fat	Moderate-weak	chip + imputation

Mechanism	Genes	Genotype (strand +, GRCh38)	Interpretation	Evidence	Trust
Oxalate	AGXT	P11L C/T + I340M A/G (common haplotype, heterozygous); G170R absent	No relevant genetic predisposition to hyperoxaluria	Moderate	chip
Purines (uric acid)	ABCG2, SLC2A9, SLC22A12	ABCG2 Q141K and Q126X absent; SLC2A9 heterozygous; SLC22A12 T/C	Low predisposition to hyperuricemia (without the most effective ABCG2 allele)	Moderate	chip + imputation
Bitter / crucifers	TAS2R38, GSTM1	TAS2R38 heterozygous in the 3 SNPs (PAV/AVI probable); Probable null GSTM1	Intermediate tester; with zero GSTM1 isothiocyanates are longer (benefit)	Moderate	chip
Phenylalanine (aspartame)	PAH	R408W, IVS12+1, IVS10-11, R261Q, R158Q, P281L, I65T, A403V, L48S absent	Without PKU (aspartame is restricted only in PKU)	Strong	chip
Galactose	GALT	N314D (Duarte) A/G heterozygous	Carrier Duarte: no clinical relevance	Moderate	chip
Salivary starch	AMY1	number of non-measurable copies	Non-determinable (CNV); SI Val15Phe can slightly reduce isomaltase	—	not determinable
FODMAP / polyols/fructose in excess	—	without validated genetic test	Not determinable by genetics	—	not determinable
Allergy IgE	HLA-DQB1, IL4R, FLG	IL4R Q576R A/G; FLG possible unconfirmed LoF; HLA rs927596 C/T	IgE food allergy not predictable by genetics (atopic context only)	Weak	partial
Mastocytes / tryptase	TPSAB1	number of non-measurable copies	Hereditary alaphatryptasemia not determinable	—	not determinable
Alpha-gal (red meat)	ABO	The probable group	Alpha-gal syndrome depends on tick bites; genetics hardly predicts	Very weak	chip + imputation
Vitamin B12 absorption/microbiome	FUT2	W154X G/A (secretor)	Secretor: typical absorption	Moderate	chip
Omega-3 / fatty acids	FADS1/2	rs174546 C/C, rs174537 G/G	Efficient conversion of ALA/LA	Moderate	chip

Colors of this table: orange/yellow : mechanism with predisposition; green = gene evaluated without predisposition; gray = non-determinable. The "informative" mechanisms (releasers of histamine, blocking DAO by tea, coffee, grapefruit, vitamin K, iron in normal portions) do not give color to any food.

3. Foods with predisposition, by category

In each category only red, orange or yellow foods appear, ordered by gravity. The last line of each category indicates how many foods were verified without genetic predisposition.

3.1 Dairy products — 42 verified

limit: 7 · surveillance: 14 · without genetic predisposition: 21

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Natal ice cream Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Condensed milk Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Blue cheese (roquefort, gorgonzola) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Feta cheese Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Sliced cheese Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak chip
Halloumi cheese Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak chip
Parmesan cheese Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Sweet rice Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Refined hydrates / high glycemic index	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; TCF7L2, FTO, MTNR1B, FABP2	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose · Intermediate glycemic response; worst at night (MTNR1B)	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate chip + imputation
Sugary yogurt with fruit Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Sugary liquid yogurt Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Milk Kefir Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Chocolate milk Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Milk cream / milk pudding Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Potatoes to beat with sugar Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Brie cheese/camembert Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Cheddar cheese Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Cured cow's cheese Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Cure goat cheese Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Buttery sheep's cheese Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Young semi-hard cheese Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Emmental cheese Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip

3.2 Starch and derivatives — 55 verified foods

limit: 2 · surveillance: 24 · without genetic predisposition: 29

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Chocolate balls Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Sweetened breakfast cereals Limit	Free sugar/fructose × liver Refined hydrates / high glycemic index Sucrose (sucrose-isomaltase) Fortified foods (iron, folic acid)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; TCF7L2, FTO, MTNR1B, FABP2; SI; HFE, MTHFR	High doses of free sugar/fructose increase liver fat in patients with 148M · Intermediate glycemic response; worst at night (MTNR1B) · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose · Iron added and synthetic folic acid: prefer natural sources	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Prefer integral versions, moderate portions, with protein or fiber and early in the day. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer versions without iron or folic acid added.	Moderate chip + imputation
Needle rice Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
White rice Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Carolina rice Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Baguette Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak chip + imputation
Rice cake/gallet Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Maria? Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Refined hydrates / high glycemic index	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; TCF7L2, FTO, MTNR1B, FABP2	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose · Intermediate glycemic response; worst at night (MTNR1B)	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate chip + imputation
Water and salt bubbles Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak chip + imputation
Croissant Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Cuscouis Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Rice flour Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Corn flour/polenta Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Corn flakes Monitor	Refined hydrates / high glycemic index Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Fortified foods (iron, folic acid) Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; HFE, MTHFR; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose · Iron added and synthetic folic acid: prefer natural sources · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer versions without iron or folic acid added. Count in daily salt limit (< 5 g).	Moderate-weak chip + imputation

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Granule Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Sugar muesli Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Papo-seco/carcass Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak chip + imputation
Pizza Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak chip + imputation
White wheat bread Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak chip + imputation
White bread Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak chip + imputation
Corn bread (corn bread) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Mixture bread (wheat and rye) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Pita/naan bread Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak chip + imputation
Gluten-free bread Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Wheat semole Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Toastes Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation

3.3 Tubercules and roots — 12 verified foods

limit: 1 · surveillance: 6 · without genetic predisposition: 5

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Potatoes of package Limit	Salt (genetic sensitivity to salt) Refined hydrates / high glycemic index	AGT, ADD1, CYP11B2, GNB3, ACE; TCF7L2, FTO, MTNR1B, FABP2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Intermediate glycemic response; worst at night (MTNR1B)	Small and infrequent portions; prefer versions with less salt. Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip
Roast potato Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Cooked potato Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Fried potato Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Cassava flour / tapioca Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Potato starch Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Potato puree Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation

3.4 Legumes and derivatives — 25 verified foods

limit: 1 · surveillance: 3 · without genetic predisposition: 21

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Miso Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Natto Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Tempeh Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Lupins Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip

3.5 Vegetables, mushrooms and algae — 67 verified foods

limit: 3 · surveillance: 13 · without genetic predisposition: 51

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Olives in brine Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Kimchi Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Pickles / cucumber in vinegar Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Avocado Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Wakame / kombu Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Canned garlic Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Aubergine Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Sauerkraut Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Coriander (fresh) Monitor	Coriander (soap flavor)	OR6A2	A slightly higher probability of finding the coriander with flavor (only flavor)	It only affects the taste; without implication for health.	Weak chip
Canned mushrooms Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Tomato concentrate Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Spinach Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Frozen spinach Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Gazpacho Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Ketchup Monitor	Food Histamine Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Count in daily salt limit (< 5 g).	Weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Roasted peppers Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip

3.6 Fruits and derivatives — 58 verified foods

limit: 7 · surveillance: 12 · without genetic predisposition: 39

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Fruit jam/sweet Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Fruit in syrup Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Marmalade Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Fruit nectar Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Package orange juice Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate chip + imputation
Apple juice Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate chip + imputation
Grape juice Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate chip + imputation
Dry plums Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate chip + imputation
Sugar-dried cranberries Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Dehydrated banana Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate chip + imputation
Dry Apricots Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate chip + imputation
Dry figs Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate chip + imputation

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Grapes Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate chip + imputation
Fruit purée (baby) Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate chip + imputation
Fruit smoothie Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate chip + imputation
Anasine juice Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate chip + imputation
Natural orange juice Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate chip + imputation
Grapefruit juice Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate chip + imputation
Dates Monitor	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate free sugar: dose-dependent hepatic effect	Moderate portion; counting in the total daily free sugar.	Moderate chip + imputation

3.7 Dry fruits and seeds — 25 verified foods

limit: 1 · surveillance: 3 · without genetic predisposition: 21

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Hazelnut and cocoa cream Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Peanuts roasted with salt Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Rice drink Monitor	Refined hydrates / high glycemic index	TCF7L2, FTO, MTNR1B, FABP2	Intermediate glycemic response; worst at night (MTNR1B)	Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip + imputation
Dry fruits with honey Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation

3.8 Meat and viscera — 32 verified foods

surveillance: 11 · without genetic predisposition: 21

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Well-done/carbonized grilled meat Monitor	Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Weakest II phase detoxification for HAP and heterocyclic amines	Avoid carbonization; marinate and cook at moderate temperature.	Weakness chip + inference (GSTM1 is CNV)

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Meat in the grill (coal) Monitor	Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Weakest II phase detoxification for HAP and heterocyclic amines	Avoid carbonization; marinate and cook at moderate temperature.	Weakness chip + inference (GSTM1 is CNV)
Heart Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak chip
Folding / tripe Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak chip
Roast chicken in the grill Monitor	Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Weakest II phase detoxification for HAP and heterocyclic amines	Avoid carbonization; marinate and cook at moderate temperature.	Weakness chip + inference (GSTM1 is CNV)
Chicken liver Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak chip
Pork liver Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak chip
Beef liver Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak chip
Beef tongue Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak chip
Molasses Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak chip
Rim Monitor	Iron heme	HFE, TMPRSS6	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Moderate portions; no iron supplementation or vitamin C in the same meal.	Weak chip

3.9 Charcutery and cured or smoked meats — 20 verified foods

limit: 17 · surveillance: 3 · without genetic predisposition: 0

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Alheira Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Smoked Bacon Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Bresaola Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Hot dog Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Meat chorizo Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Blood chorizo / black pudding Limit	Salt (genetic sensitivity to salt) Food Histamine Iron heme	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; HFE, TMPRSS6	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk)	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portions; no iron supplementation or vitamin C in the same meal.	Moderate-weak chip
Farinheira Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Tongue Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Mortadella Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak chip
Cured pork loin sausage / curated loin Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Pancetta Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Pastrami / smoked meat Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Curated ham Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Salami Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Salpicon Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Smoked sausages Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Smoked meat Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Braid Monitor	Salt (genetic sensitivity to salt) Fortified foods (iron, folic acid)	AGT, ADD1, CYP11B2, GNB3, ACE; HFE, MTHFR	Moderate-high salt: counting daily · Iron added and synthetic folic acid: prefer natural sources	Count in daily salt limit (< 5 g). Prefer versions without iron or folic acid added.	Moderate-weak chip
Turkey ham Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Liver paw Monitor	Iron heme Salt (genetic sensitivity to salt) Food Histamine	HFE, TMPRSS6; AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Very rich in heme iron; H63D carrier with slightly higher absorption (low overload risk) · Moderate-high salt: counting daily · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Moderate portions; no iron supplementation or vitamin C in the same meal. Count in daily salt limit (< 5 g). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip

3.10 Fish — 42 verified foods

limit: 4 · surveillance: 12 · without genetic predisposition: 26

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Canned anchovy/salted anchovy Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Smoked herb Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Salted dry fish (others) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Smoked salmon Limit	Salt (genetic sensitivity to salt) Food Histamine Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Fresh herb Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Canned tuna Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Fresh tuna Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Dry salted cod (wet) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Horse Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Fish douradines Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Swordfish Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Fish eggs Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Roasted sardines (in the coal) Monitor	Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Weakest II phase detoxification for HAP and heterocyclic amines	Avoid carbonization; marinate and cook at moderate temperature.	Weakness chip + inference (GSTM1 is CNV)
Canned sardine Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Surimi / Delightful Sea Monitor	Salt (genetic sensitivity to salt) Sucrose (sucrose-isomaltase)	AGT, ADD1, CYP11B2, GNB3, ACE; SI	Moderate-high salt: counting daily · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Count in daily salt limit (< 5 g). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate-weak chip
Sushi / sashimi Monitor	Salt (genetic sensitivity to salt) Refined hydrates / high glycemic index	AGT, ADD1, CYP11B2, GNB3, ACE; TCF7L2, FTO, MTNR1B, FABP2	Moderate-high salt: counting daily · Intermediate glycemic response; worst at night (MTNR1B)	Count in daily salt limit (< 5 g). Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip

3.11 Shellfish and mollusks — 20 verified foods

limit: 1 · surveillance: 1 · without genetic predisposition: 18

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Dry shrimp Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Canned shellfish Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip

3.12 Eggs — 7 verified foods

no food with predisposition · without genetic predisposition: 7

3.13 Fats and oils — 19 verified foods

no food with predisposition · without genetic predisposition: 19

3.14 Non-alcoholic beverages (coffee, tea, cocoa, juices, soft drinks) — 35 verified foods

limit: 13 · surveillance: 14 · without genetic predisposition: 8

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Guaraná beverage Limit	Free sugar/fructose x liver Coffee Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; CYP1A2, ADORA2A; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Moderate caffeine: attention to total dose and time · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Count total caffeine of the day; avoid in the afternoon and at night. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Energy drink Limit	Coffee Free sugar/fructose x liver Sucrose (sucrose-isomaltase)	CYP1A2, ADORA2A; PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p80): dose and time dependent effect · High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Up to ~200 mg of caffeine per day and only in the morning. Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip
Sugar-free energy drink Limit	Coffee	CYP1A2, ADORA2A	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p80): dose and time dependent effect	Up to ~200 mg of caffeine per day and only in the morning.	Moderate chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Smoked or boiled coffee (not filtered) Limit	Coffee	CYP1A2, ADORA2A	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p80): dose and time dependent effect	Up to ~200 mg of caffeine per day and only in the morning.	Moderate chip
Expressed coffee Limit	Coffee	CYP1A2, ADORA2A	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p80): dose and time dependent effect	Up to ~200 mg of caffeine per day and only in the morning.	Moderate chip
Filtered coffee Limit	Coffee	CYP1A2, ADORA2A	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p80): dose and time dependent effect	Up to ~200 mg of caffeine per day and only in the morning.	Moderate chip
Sugar iced tea Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Tea matcha Limit	Coffee	CYP1A2, ADORA2A	Normal to intermediate elimination and less sensitive ADORA2A; the limit comes from sleep (insomnia p80): dose and time dependent effect	Up to ~200 mg of caffeine per day and only in the morning.	Moderate chip
Gas / lemonade Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Powdered chocolate milk (chocolate) Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Glue refrigerator Limit	Free sugar/fructose × liver Coffee Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; CYP1A2, ADORA2A; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Moderate caffeine: attention to total dose and time · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Count total caffeine of the day; avoid in the afternoon and at night. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Orange / lemon Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Tonic water (quinine) Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Isotonic beverage Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Cocoa powder Monitor	Coffee Food Histamine	CYP1A2, ADORA2A; AOC1 (DAO), HNMT	Moderate caffeine: attention to total dose and time · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Count total caffeine of the day; avoid in the afternoon and at night. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate chip
Instant coffee Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate chip
Cappuccino Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate chip
Beer without alcohol Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Hot chocolate Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Food Histamine	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; AOC1 (DAO), HNMT	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate chip + imputation
Mate tea Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate chip
Tea oolong Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate chip
Black tea Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate chip
Green tea Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate chip
Galon / coffee with milk Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate chip
Infusion of liquorice Monitor	Liquorice (glycyrrhizal)	CYP11B2, AGT, ADD1	Glycyrrhizin retains sodium; more relevant with salt sensitivity	Avoid the regular consumption of licorice.	Weak chip
Kombucha Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Glue refrigerator without sugar Monitor	Coffee	CYP1A2, ADORA2A	Moderate caffeine: attention to total dose and time	Count total caffeine of the day; avoid in the afternoon and at night.	Moderate chip

3.15 Alcoholic beverages — 16 verified foods

limit: 16 · without genetic predisposition: 0

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Brandy / bagasse Limit	Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Zero alcohol.	Moderate-strong chip + imputation
Beer Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong chip + imputation
Sweetened cocktails Limit	Alcohol Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Normal acute tolerance, but higher hepatic susceptibility to alcohol · High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Zero alcohol. Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate-strong chip + imputation
Foaming / cleanliness Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong chip + imputation
Gin Limit	Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Zero alcohol.	Moderate-strong chip + imputation
Liqueurs (gink, bitter almond) Limit	Alcohol Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Normal acute tolerance, but higher hepatic susceptibility to alcohol · High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Zero alcohol. Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate-strong chip + imputation
Rum Limit	Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Zero alcohol.	Moderate-strong chip + imputation
Sacqué Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong chip + imputation
Sidra Limit	Alcohol Food Histamine Free sugar/fructose × liver	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT; PNPLA3, MBOAT7, SERPINA1, HSD17B13	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate free sugar: dose-dependent hepatic effect	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portion; counting in the total daily free sugar.	Moderate-strong chip + imputation
White wine Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong chip + imputation

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Rosé wine Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong chip + imputation
Red wine Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong chip + imputation
Green wine Limit	Alcohol Food Histamine	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-strong chip + imputation
Generous / fortified wines Limit	Alcohol Food Histamine Free sugar/fructose x liver	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7; AOC1 (DAO), HNMT; PNPLA3, MBOAT7, SERPINA1, HSD17B13	Normal acute tolerance, but higher hepatic susceptibility to alcohol · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate free sugar: dose-dependent hepatic effect	Zero alcohol. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portion; counting in the total daily free sugar.	Moderate-strong chip + imputation
Vodka Limit	Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Zero alcohol.	Moderate-strong chip + imputation
Scotch Limit	Alcohol	ALDH2, ADH1B, PNPLA3, SERPINA1, MBOAT7	Normal acute tolerance, but higher hepatic susceptibility to alcohol	Zero alcohol.	Moderate-strong chip + imputation

3.16 Fermented and preserved fermented — 13 verified foods

limit: 7 · surveillance: 5 · without genetic predisposition: 1

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Canned sauerkraut Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Beer yeast / yeast extract Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Fermented pepper mass Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Fish sauce (nam pla) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Soybean sauce Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Fermented pickles (lacto-fermented) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Garlic and pepper fermented seasoning (for meat) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Water Kefir Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Mother-mass bread Monitor	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Moderate-high salt: counting daily · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Count in daily salt limit (< 5 g). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Balsamic wine Monitor	Food Histamine Free sugar/fructose × liver	AOC1 (DAO), HNMT; PNPLA3, MBOAT7, SERPINA1, HSD17B13	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate free sugar: dose-dependent hepatic effect	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portion; counting in the total daily free sugar.	Weak chip
Cider vinegar Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip
Wine wine Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip

3.17 Sugars, sweeteners and sweets — 30 verified foods

limit: 20 · surveillance: 2 · without genetic predisposition: 8

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Sugarcane (liquorice-sweet) Limit	Free sugar/fructose × liver Liquorice (glycyrrhizal) Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; CYP11B2, AGT, ADD1; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Glycyrrhizin retains sodium; more relevant with salt sensitivity · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Avoid the regular consumption of licorice. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Sweet cotton Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Yellow/masked sugar Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
White sugar Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Whole sugarcane sugar (grape) Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Bombos Limit	Free sugar/fructose × liver Food Histamine Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; AOC1 (DAO), HNMT; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
White chocolate Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Milk chocolate Limit	Free sugar/fructose × liver Food Histamine Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; AOC1 (DAO), HNMT; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Milk sweet Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Eggs/mole eggs Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Powder fructose Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate chip + imputation
Sugared gelatin Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Marmalade / quince jelly Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Mel Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate chip + imputation
Sugarcane molasses Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Nougat / roast Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Sweeteners/gums Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Agave syrup Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate chip + imputation
Glucose-fructose syrup (HFCS) Limit	Free sugar/fructose × liver	PNPLA3, MBOAT7, SERPINA1, HSD17B13	High doses of free sugar/fructose increase liver fat in patients with 148M	Limiting free sugars and sugary drinks (the whole fruit is not the problem).	Moderate chip + imputation
Maple syrup Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Black chocolate 70% Monitor	Coffee Food Histamine	CYP1A2, ADORA2A; AOC1 (DAO), HNMT	Moderate caffeine: attention to total dose and time · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Count total caffeine of the day; avoid in the afternoon and at night. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate chip
Chocolate mousse Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Food Histamine	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; AOC1 (DAO), HNMT	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate chip + imputation

3.18 Spices, herbs, condiments and sauces — 45 verified foods

limit: 7 · surveillance: 12 · without genetic predisposition: 26

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Concentrated broth (cube) Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Meat extract Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Chilli mass Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Worcestershire Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Barbecue sauce Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase) Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI; AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose · Moderate-high salt: counting daily · Weakest II phase detoxification for HAP and heterocyclic amines	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose. Count in daily salt limit (< 5 g). Avoid carbonization; marinate and cook at moderate temperature.	Moderate chip + imputation
Cooking salt Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak chip
Sea salt / salt flower Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak chip
Dry coriander seeds Monitor	Coriander (soap flavor)	OR6A2	A slightly higher probability of finding the coriander with flavor (only flavor)	It only affects the taste; without implication for health.	Weak chip
Monosodium glutamate Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Pyri-pyri sauce in oil Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Soybean sauce reduced in salt Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Industrial tomato sauce Monitor	Food Histamine Free sugar/fructose × liver Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; PNPLA3, MBOAT7, SERPINA1, HSD17B13; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate free sugar: dose-dependent hepatic effect · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Moderate portion; counting in the total daily free sugar. Count in daily salt limit (< 5 g).	Weak chip
Hot sauce (tabasco, sriracha) Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Mustard (wet) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Smoked paprika Monitor	Smoked/carbonized meat (HAP, heterocyclic amines)	GSTM1, GSTA1, NAT2	Weakest II phase detoxification for HAP and heterocyclic amines	Avoid carbonization; marinate and cook at moderate temperature.	Weakness chip + inference (GSTM1 is CNV)
Pestus Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Liquorice root (cattlewood) Monitor	Liquorice (glycyrrhizal)	CYP11B2, AGT, ADD1	Glycyrrhizin retains sodium; more relevant with salt sensitivity	Avoid the regular consumption of licorice.	Weak chip
Green parsley / chimichurri Monitor	Coriander (soap flavor)	OR6A2	A slightly higher probability of finding the coriander with flavor (only flavor)	It only affects the taste; without implication for health.	Weak chip
Vinagret Monitor	Food Histamine	AOC1 (DAO), HNMT	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Weak chip

3.19 Pastry, desserts and sweet snacks — 24 verified foods

limit: 19 · surveillance: 4 · without genetic predisposition: 1

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Camel Baba Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrose slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Butterball Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrose slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Chocolate cake Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrose slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Fried cake filled with cream Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrose slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Bolo-Rei Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrose slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Cheesecake Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrose slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Donut Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Burned cream Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Muffin Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Palmier Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Pastel de nata Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Pudding flan Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Ló bread Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Queijada Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Horseradish Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Dreams / children Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Fruit ice cream Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Tiramisu Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Pillow / sweet leaf Limit	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	High doses of free sugar/fructose increase liver fat in patients with 148M · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Limiting free sugars and sugary drinks (the whole fruit is not the problem). If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Cereal bar Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Oats Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Apple pie Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation
Waffles / sweet crepes Monitor	Free sugar/fructose × liver Sucrose (sucrose-isomaltase)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; SI	Moderate free sugar: dose-dependent hepatic effect · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose	Moderate portion; counting in the total daily free sugar. If there are gases or diarrhea with sweets, try to reduce sucrose.	Moderate chip + imputation

3.20 Ready meals, fast food and processed salty foods — 39 verified

limit: 14 · surveillance: 23 · without genetic predisposition: 2

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Sugar-tuffed rice Limit	Free sugar/fructose × liver Refined hydrates / high glycemic index Sucrose (sucrose-isomaltase) Fortified foods (iron, folic acid)	PNPLA3, MBOAT7, SERPINA1, HSD17B13; TCF7L2, FTO, MTNR1B, FABP2; SI; HFE, MTHFR	High doses of free sugar/fructose increase liver fat in patients with 148M · Intermediate glycemic response; worst at night (MTNR1B) · Activity of sucrase slightly lower; possible digestive discomfort with much sucrose · Iron added and synthetic folic acid: prefer natural sources	Limiting free sugars and sugary drinks (the whole fruit is not the problem). Prefer integral versions, moderate portions, with protein or fiber and early in the day. If there are gases or diarrhea with sweets, try to reduce sucrose. Prefer versions without iron or folic acid added.	Moderate chip + imputation
Meat and sausage cooking Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Bean Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
France Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Fast food burger Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak chip
Kebab/shawarma Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak chip
Instant mass (noodles) Limit	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt	Small and infrequent portions; prefer versions with less salt.	Moderate-weak chip
Pepperoni pizza Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Pretzels Limit	Salt (genetic sensitivity to salt) Refined hydrates / high glycemic index	AGT, ADD1, CYP11B2, GNB3, ACE; TCF7L2, FTO, MTNR1B, FABP2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Intermediate glycemic response; worst at night (MTNR1B)	Small and infrequent portions; prefer versions with less salt. Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip
Bread with sausage Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Ham sandwich Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Salted corn snacks Limit	Salt (genetic sensitivity to salt) Refined hydrates / high glycemic index	AGT, ADD1, CYP11B2, GNB3, ACE; TCF7L2, FTO, MTNR1B, FABP2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Intermediate glycemic response; worst at night (MTNR1B)	Small and infrequent portions; prefer versions with less salt. Prefer integral versions, moderate portions, with protein or fiber and early in the day.	Moderate-weak chip
Instantaneous soup in package Limit	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Small and infrequent portions; prefer versions with less salt. Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Strips of bacon / towers Limit	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Sensitivity to increased salt (polygenic): the tension tends to rise more with much salt · Weakest II phase detoxification for HAP and heterocyclic amines	Small and infrequent portions; prefer versions with less salt. Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip
Rice of shellfish Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Duck rice Monitor	Salt (genetic sensitivity to salt) Smoked/carbonized meat (HAP, heterocyclic amines)	AGT, ADD1, CYP11B2, GNB3, ACE; GSTM1, GSTA1, NAT2	Moderate-high salt: counting daily · Weakest II phase detoxification for HAP and heterocyclic amines	Count in daily salt limit (< 5 g). Avoid carbonization; marinate and cook at moderate temperature.	Moderate-weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Cod with creams Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Bras Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Fast food potatoes Monitor	Refined hydrates / high glycemic index Salt (genetic sensitivity to salt)	TCF7L2, FTO, MTNR1B, FABP2; AGT, ADD1, CYP11B2, GNB3, ACE	Intermediate glycemic response; worst at night (MTNR1B) · Moderate-high salt: counting daily	Prefer integral versions, moderate portions, with protein or fiber and early in the day. Count in daily salt limit (< 5 g).	Moderate-weak chip + imputation
Biphane (pig sandwich) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Burrito / batos Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Fish caulder Monitor	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Moderate-high salt: counting daily · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Count in daily salt limit (< 5 g). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Curry with rice Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Salted pastries/pastries Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Beans in can with sauce Monitor	Salt (genetic sensitivity to salt) Free sugar/fructose × liver	AGT, ADD1, CYP11B2, GNB3, ACE; PNPLA3, MBOAT7, SERPINA1, HSD17B13	Moderate-high salt: counting daily · Moderate free sugar: dose-dependent hepatic effect	Count in daily salt limit (< 5 g). Moderate portion; counting in the total daily free sugar.	Moderate-weak chip
Chicken pyri Monitor	Smoked/carbonized meat (HAP, heterocyclic amines) Salt (genetic sensitivity to salt)	GSTM1, GSTA1, NAT2; AGT, ADD1, CYP11B2, GNB3, ACE	Weakest II phase detoxification for HAP and heterocyclic amines · Moderate-high salt: counting daily	Avoid carbonization; marinate and cook at moderate temperature. Count in daily salt limit (< 5 g).	Weakness chip + inference (GSTM1 is CNV)
Infant formula / dairy papayas Monitor	Fortified foods (iron, folic acid)	HFE, MTHFR	Iron added and synthetic folic acid: prefer natural sources	Prefer versions without iron or folic acid added.	Weak chip
Lasagne Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Chicken nuggets Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Paella Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Pizza margherita Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Prego (sands of steak) Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip

Food	Mechanism(s)	Genes	Predicted response	Suggestion (not applied)	Evidence · trust
Frozen ready meal Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Rissols / croquettes / cod pastes Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip
Canned tuna salad Monitor	Food Histamine Salt (genetic sensitivity to salt)	AOC1 (DAO), HNMT; AGT, ADD1, CYP11B2, GNB3, ACE	Mildly reduced histamine degradation: possible sensitivity with high doses of histamine · Moderate-high salt: counting daily	Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor. Count in daily salt limit (< 5 g).	Weak chip
Cheese sandwich Monitor	Salt (genetic sensitivity to salt) Food Histamine	AGT, ADD1, CYP11B2, GNB3, ACE; AOC1 (DAO), HNMT	Moderate-high salt: counting daily · Mildly reduced histamine degradation: possible sensitivity with high doses of histamine	Count in daily salt limit (< 5 g). Prefer fresh; if they appear flushing, headache or congestion after these foods, review with the doctor.	Moderate-weak chip
Mixed toast Monitor	Salt (genetic sensitivity to salt)	AGT, ADD1, CYP11B2, GNB3, ACE	Moderate-high salt: counting daily	Count in daily salt limit (< 5 g).	Moderate-weak chip

Abstract: 324 of the 626 verified foods have no genetic predisposition and are not listed.

4. Limitations

- **Genetics only explains part of the intolerances.** Most reactions to food depend on the intestine, microbiome, dose, preparation and other non-genetic factors.
- **IgE allergies** (brown, dried fruits, shellfish, fish, egg, milk, sesame, etc.) are not predictable by genetics: IL4R Q576R heterozygous, the possible loss of filaggrin function (unconfirmed) and the loci HLA give only atopic context. The dietary IgG tests do not serve to decide evictions.
- **FODMAP, polyols and fructose malabsorption** (sorbitol, xylitol, fructose in excess, legumes) and the sensitivity to non-celiac gluten do not have a validated genetic test; they are non-determinable.
- **Chip + imputation, no sequencing.** Only the variants measured or imputed with $DR2 \geq 0.8$ are seen; rare ($< 0.1\%$) and new variants are not seen. A result "without predisposition" means "without the variants tested", not "without variants". Not measurable: AMY1 (amylase, CNV), TPSAB1 (trypsinase, CNV), GSTM1/GSTT1 (inferred), repetitions and high resolution HLA.
- **Dose effects:** orange and yellow indicate dose- and frequency-dependent effects (sugar, salt, caffeine, alcohol, histamine), neither allergy nor mandatory intolerance. Colors assume usual portions.
- **Evidence:** histamine (DAO/HNMT), sucrase (SI Val15Phe), coriander, iron (H63D) and smoked (GSTM1/GSTA1/NAT2) are based on association studies with small effects. Lactase, HLA-DQ celiac, ALDOB, G6PD and PAH have strong evidence.
- **List of foods:** results from a selection based on statistics and composition tables; the real composition (salt, sugar, histamine) varies by brand, revenue and maturation.
- **It is not a diagnosis.** If there are symptoms, confirm with clinical tests: hydrogen respiratory test (lactose, fructose, sucrose), tTG-IgA with total IgA (celiac disease, still eating gluten), allergy tests IgE with the allergologist and, for histamine, medical evaluation with a food diary. All indications are suggestion (not applied).

5. Sources and methods

- List of foods: products from FAO Food Balance Sheets (FAOSTAT, 2010–2023), tables of food and food composition, dishes and processed frequent consumption worldwide; manual deduplication and classification in 21 categories.
- Genotypes: genotyping chip (GRCh38) as primary source; Beagle 5.5 imputation with the full genome reference panel 30x (only $DR2 \geq 0.8$); HLA tags calibrated in the reference population.
- ClinVar (28-09-2026 version): Scanning of pathogenic/probably pathogenic variants in ~130 food metabolism genes (ALDOB, SI, LCT, PAH, AGXT, GALT, G6PD, AOC1, HNMT, HFE, urea cycle, among others).
- Background: persistence of lactase (Enattah 2002); HLA-DQ and celiac disease (Monsuur 2008; ESPGHAN); ALDOB (Cross 1990; Santer 2005); SI and IBS (Henström 2018); DAO/HNMT (Maintz 2011; García-Martin 2015); CYP1A2/ADORA2A (Cornelis 2006; Rétey 2007; Cornelis 2011); salt (consensus AHA on sensitivity to salt 2016); PNPLA3 × sugar and alcohol (Davis 2010; Sevastianova 2011; Buch 2015); APOA2 (Corella 2009); GSTM1/NAT2 and cooked meat (IARC 2018); OR6A2 (Eriksson 2012); G6PD (OMS); PAH (ClinVar).